

Question number	Answer	Additional guidance	
1(a)	A description that includes two of the following points: - calcium ions bind to troponin (1) - causing tropomyosin to move (1) - allowing the myosin (head) to bind (1)	ACCEPT change in shape ACCEPT exposing the myosin binding site / allowing actomyosin bridges to form	(2)

Question number	Answer	Additional guidance	
1(b)	An explanation that includes the following points: - many mitochondria allows more (aerobic) respiration(1) - many myoglobin molecules allows greater (store)of oxygen (1) - larger capillary network ensures good supply of {oxygen/ glucose} (1) - slow sustained contraction allows longer periods ofexercise / race for longer(1)	ACCEPT more ATP produced ACCEPT in context endurance events / decrease in {fatigue / tiredness}	(4)

Question number	Answer	Additional guidance	
1(c)	An answer that includes the following points: • exercise increases the rate of heat loss (1) • (eventually) the rate of heat loss reaches its maximum (1)	ACCEPT the rate of heat loss levels off	(2)